

Depth Reduction in Light-Field Head-Mounted Displays by Generating Intermediate Images as Virtual Images

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ABSTRACT

Light-field head-mounted displays (HMDs) can resolve vergence–accommodation conflicts, which may cause visual discomfort and fatigue. However, light-field HMDs have a narrow field of view (FOV), owing to their small display size and optical configuration. We increased the FOV by adopting a large display and shifting the elemental image from the back of the corresponding microlens. In this study, we proposed a method to generate intermediate images as virtual images to reduce the distance between the microlens array and the eyepiece. We also designed a microlens array suitable for the proposed method and successfully reduced the device depth by 44%.

Keywords: Light field, head mounted display, virtual reality.

Index Terms: Computer Graphics Methodologies—Computer Graphics—Virtual Reality; Computer Graphics Methodologies—Rendering—Ray Tracing

1 INTRODUCTION

Head-mounted displays (HMDs) that use light-field technology generate natural 3D images by reproducing the light rays that reach the eye from an object. In principle, these displays overcome the convergence–accommodation conflict and are expected to reduce eye strain and visual fatigue.

Studies that use microlens arrays (MLAs) or pinhole arrays have reported a thin, lightweight light-field HMD for VR [1, 2], and optical see-through light-field augmented-reality (AR) glasses using an MLA and a free-form prism as the eyepiece [3]. In these studies, microdisplays with small screens (i.e., with a diagonal size of less than two inches) are typically used as display elements because of their light weight and low power consumption. However, 3D-image reconstruction and a wide field of view (FOV) require a large number of pixels in the light-field HMD. We previously employed the 5.5-inch 4K liquid crystal display (LCD) panel that is used in smartphones and achieved a larger FOV by making the pitch of the elemental image larger than that of the MLA [4]. However, in the current configuration, an intermediate image is formed between the MLA and the eyepiece, which increases the device depth. Because HMDs for VR generally place the display and optical elements in front of the face, the device depth should be small.

In this study, we proposed a method to reduce the distance between the MLA and the eyepiece by changing the formation position of the intermediate image and verified the principle through experiments. We also designed and applied a new MLA suitable for the proposed method and demonstrated its effectiveness.

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2 INTERMEDIATE IMAGE-DISPLAY METHOD

Light-field HMDs consist of a display, an MLA, and an eyepiece. In the conventional method, the MLA is arranged farther from the display panel than its focal length, and the intermediate image is formed as a real image. The optical conjugate plane of the display to the MLA is called the intermediate image plane, and the position of the intermediate image plane is determined by the following equation:

$$b = \frac{af_{\text{array}}}{a - f_{\text{array}}}, \quad (1)$$

where a is the distance between the display and MLA, b is the distance between the MLA and the intermediate image plane, and f_{array} is the focal length of the MLA.

In the proposed method, the intermediate image is formed as a virtual image by ensuring that the distance between the display and the MLA is shorter than the focal length of the MLA. In this case, the position of the intermediate image plane is behind the display (Fig. 1), and the distance between the MLA and intermediate image plane b' is specifically determined by the following equation:

$$b' = \frac{af_{\text{array}}}{a + f_{\text{array}}}. \quad (2)$$

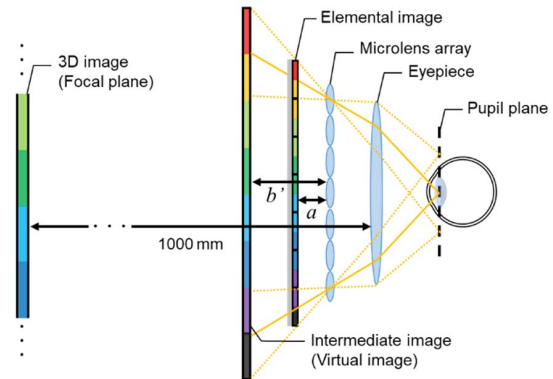


Figure 1: Configuration of light-field HMD. The intermediate image is formed as a virtual image.

The position of the eyepiece is determined by the positions of the intermediate image plane and the 3D image. By forming the intermediate image as a virtual image at the back of the display, the distance between the MLA and the eyepiece can be decreased, thereby reducing the device depth.

3 SIMULATION AND EXPERIMENT

We conducted simulations and experiments to determine whether 3D images could be displayed correctly using the proposed method. A 5.5-inch display with a 4K resolution was used as the display, and a Fresnel lens with a 100-mm focal length was used

as the eyepiece. The specifications of the MLA are presented in Table 1. Table 2 shows the arrangement of the display and optical elements under the same conditions, where only the arrangement changed, and the newly designed MLA was applied. We found that the device size (distance from the display to the eyepiece) can be reduced by 40% with the same optical elements and by 44% with the newly designed MLA, compared to the conventional method.

Red, green, and blue Stanford dragons were used as 3D objects and placed at distances of 700, 1000, and 1300 mm, respectively, from the eyepiece. Elemental images were generated by tracking the rays from each pixel of the display to the 3D objects.

The optical simulation was performed using the OpticStudio (Zemax, LLC) optical-design software. The elements were arranged in the software, as shown in Table 2, and a lens and a detector, corresponding to the crystalline lens and the retina, were added. The MLA, eyepiece, and crystalline lens were all ideal lenses with no optical aberrations. Using the generated elemental images as a surface light source, ray tracing was performed by randomizing the starting position and direction to generate a retinal image. In the experiment, the elements were placed on an optical bench, as shown in Table 2, and a camera was placed at the pupil plane to take photographs.

Table 1: MLA specifications.

	Conventional ^[4]	Newly designed
Pitch	1.38 mm	1.00 mm
Focal length	8.85 mm	15.0 mm
Microlens shape	Round	Square
Arrangement	Delta	Orthogonal

Table 2: Distance between the display and optical elements.

Condition	Adopt the conventional MLA		Adopt the newly designed MLA
	Conventional	Proposed	Proposed
Method			
Display – MLA	12.2 mm	5.5 mm	9.3 mm
MLA – Eyepiece	123.2 mm	76.3 mm	66.1 mm
Eyepiece – Pupil Plane	57.0 mm	57.0 mm	42.0 mm
Device size	135.4 mm	81.8 mm	75.4 mm

Fig. 2 shows the simulated and experimental results. We confirmed the formation of 3D images by the proposed method through a simulated retinal image and a photograph of the prototype.

In the following, we compare the differences when the same MLA is used in both the conventional and proposed methods (red and green dotted lines in Fig. 2). First, for the simulated retinal image, the display area was extended using the proposed method. The black areas observed in the conventional method can be attributed to the fact that the number of display pixels in the simulation was limited to 1080×1080 pixels. This indicates that the FOV can be expanded by adopting the proposed method, when using a display with the same number of pixels.

Second, the simulated retinal image of the proposed method exhibited uneven luminance along the MLA arrangement. In the proposed method, the distance between the display and the MLA was smaller, resulting in a larger eyebox. The eyebox is the area in which the reconstructed 3D image can be seen normally without disturbance in the pupil plane. In the proposed method, the eyebox was larger and the pupil ratio in the eyebox was reduced; thus, fewer light rays reached the eye, causing uneven luminance. In contrast, no uneven luminance was observed in the photograph taken by the prototype, which can be attributed to the fact that sufficient light quantity was obtained from the display.

Third, the simulated retinal image of the proposed method exhibited the collapse of the square-wave texture at a depth of 700

mm (above the red dragon) or 1300 mm (above the blue dragon) from the eyepiece. This is because the distance between the display and the MLA is smaller in the proposed method. This increases the numerical aperture of the MLA, which decreases the depth of field.

The blue dotted line in Fig. 2 shows the simulated retinal image and the photograph obtained using the newly designed MLA. In comparing the newly designed MLA with the conventional MLA in the proposed method, problems such as uneven luminance and square-wave texture disorder were eliminated. These problems can be alleviated by using an MLA with a longer focal length and optically designing the eyebox to be comparable to that of conventional methods.

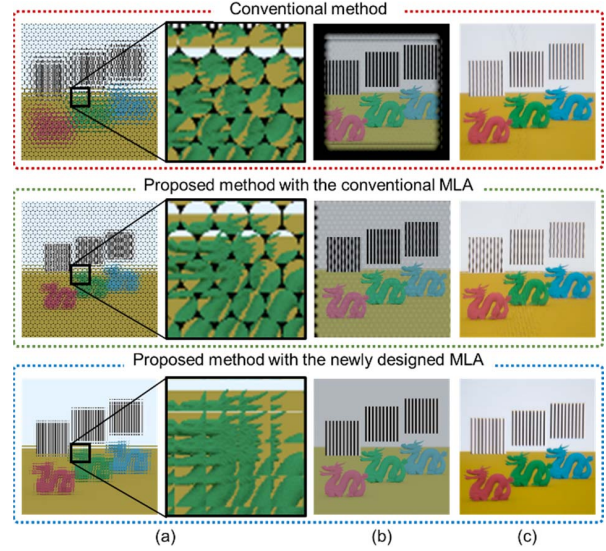


Figure 2: Simulated and experimental results. (a) Elemental images, (b) simulated retinal images, and (c) photographs.

4 CONCLUSION

We proposed a method to form an intermediate image as a virtual image to reduce the depth of light-field HMDs. By applying the proposed method, we showed that the depth of the light-field HMD for VR could be reduced by 44% and achieved a depth size close to that of common binocular HMDs. We will demonstrate the feasibility of the proposed method and execute subjective evaluation of the image quality by prototyping the HMD that can be worn on the head and supports head tracking.

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